

OUTRAM PARK: A ROADMAP

An open, unified, human-verified multiphysics toolkit for anything nuclear

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SECTION 1

The Vision

OUTRAM PARK IN ONE SENTENCE

A **one-stop, open-source multiphysics toolkit** where researchers collaborate on **anything nuclear**, on one common, unified codebase that is **drafted with AI and verified by humans**.

- ⦿ **Open-source Unified TRAnsient Multi-Physics Advanced Reactor simulation Kit.**
- ⦿ Neutronics, thermal hydraulics and CFD, fuel and structures, materials, chemistry and source term in **one Rust workspace**, with one API style and one V&V discipline.
- ⦿ **Long-run claim:** OUTRAM PARK can model *any* Gen IV reactor.
- ⦿ **Endgame:** a self-sustaining community, as OpenFOAM has.

THE PROBLEM IT ADDRESSES

Nuclear multiphysics today is **fragmented**:

- ◉ Each phenomenon lives in a different code, with its own language, input format, licence and access regime.
- ◉ Coupling them is a research project in itself, and is often redone by every group.
- ◉ Many reference codes are export-controlled or closed, which is hard on students, small groups and new nuclear countries.
- ◉ Building input decks for system codes such as MELCOR or ASTEC is slow, so few scenarios are ever explored.

OUTRAM PARK's answer: port the established open codes into one workspace, measure each port against its upstream, and put everything behind a common API that humans (and AI agents) can actually use.

WHY NOW: AI DRAFTS, HUMANS VERIFY

AI makes the breadth possible

- ⦿ Translating OpenFOAM, OpenMC, NJOY2016, CoolProp, OFFBEAT, ... into one language was never affordable for one group.
- ⦿ AI drafting has produced $\sim 1.27\text{M}$ lines across the workspace in under two years.

Humans make it trustworthy

- ⦿ Everything is **unverified until validated**.
- ⦿ AI-generated V&V is a *screening gate*, not a verdict.
- ⦿ A code is **ready** only after humans have studied it and reproduced its cases themselves.

The two halves need each other. AI without human V&V is untrustworthy; human effort without AI drafting is too slow to cover a whole reactor.

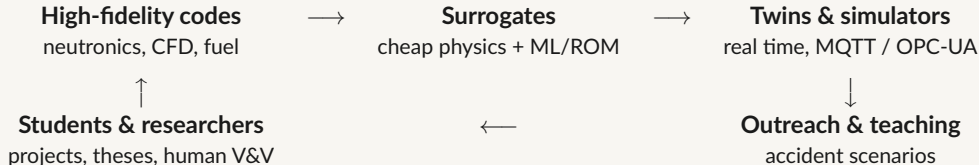
SECTION 2

Who It Serves

FOUR GROUPS, ONE CODEBASE

Group	What they do with it	Priority
Researchers	High-fidelity study of phenomena: neutronics, boiling, precursor drift, double heterogeneity, Kinchin–Pease damage, stress corrosion cracking	Now
Digital twins	Real-time simulators on cheap physics models and ML/ROM surrogates; optionally coupled to integral effects tests	Now
Students & public	Outreach simulators with accident scenarios; undergrad and postgrad projects and theses in the source code	Now
Safety analysts	Cheap accident-scenario exploration ahead of MELCOR / ASTEC	When SEMBAWANG is reviewed

HOW THE FOUR GROUPS FEED EACH OTHER



- ◉ High-fidelity runs produce the **training data** for ML/ROM surrogates.
- ◉ Surrogates and cheap models power the **twins** and the **outreach simulators**.
- ◉ Outreach draws in **students**, whose projects on the source code become **human V&V**, which makes the high-fidelity codes ready.

RESEARCHERS AND STUDENTS: INTO THE SOURCE

- ◉ **Researchers** study phenomena at high fidelity, on the same codebase as their collaborators, rather than coupling five codes by hand.
- ◉ **Undergraduates and postgraduates** work directly in the source code through final-year projects, course projects and theses.
- ◉ Their hand-written cases *are* human V&V: every student project that reproduces a benchmark through the API moves a crate toward ready.
- ◉ API friction they hit is recorded and fixed. The API is designed to be usable by a newcomer (and by a small AI model) without reading the source.

For **public outreach**, prepared simulators with scripted accident scenarios fit better than source code. These build on the teaching decks already in presentations/.

DIGITAL TWINS: TWO KINDS OF SURROGATE

Cheap physics models

- ◉ Point kinetics (teh-o-prke), 1-D thermal hydraulics (tuas), steam cycles (tampines).
- ◉ Run in real time today: the FHR, HTGR and CIET simulators.

Deployment path:

- 1 The twin is a **real-time simulator** first.
- 2 If desired, it **connects to an integral effects test** facility over MQTT and OPC-UA. CIET v2 already ships an OPC-UA server.

Offline research and education only. Never connected to operating plant or safety systems.

ML / ROM surrogates

- ◉ Trained on OUTRAM PARK's own high-fidelity codes.
- ◉ Bring higher-fidelity physics to real time.

SEMBAWANG (Severe-accident Evolution and Melt Behaviour Analysis Workbench for Advanced Nuclear Geometries) is the *cheap* model.

- 1 AI + SEMBAWANG **explore many accident scenarios** quickly.
 - 2 **Narrow down** to the scenarios that actually need a full assessment.
 - 3 **Verify the parameters** in SEMBAWANG models, which are quick to build.
 - 4 **Compile the information** the MELCOR / ASTEC input decks need.
 - 5 Engineers build the decks; **real assessment begins** in MELCOR / ASTEC.
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Status: TRISO source term implemented (via `boon-lay`); accident progression not yet. It orchestrates the offsite chain: `changi` (dispersion) and `redhill` (ground transport). No dose is computed. Under AI drafting, needs review; this group waits until it is ready.

SECTION 3

What Exists Today

THE WORKSPACE TODAY

outram-park-backend

- ⦿ **44 crates**; ~1.27M lines of Rust (measured at 38 crates, 15 Sep 2026).
- ⦿ Ports of OpenFOAM, OpenMC, NJOY2016, CoolProp, OFFBEAT, LIGGGHTS, Moltres, PFLOTRAN, DWSIM, ...
- ⦿ **12 crates declared mature** (15 Sep 2026), each with a written bar and evidence class.
- ⦿ Every crate README carries a **maintainer sign-off tracker**. None is signed off yet.

Per-module detail: *Where OUTRAM PARK Stands*, the status deck of 15 Sep 2026.

outram-park (this repo)

- ⦿ The front end: desktop GUI, a CLI/TUI that runs on Termux, and a Python wheel over the whole backend.
- ⦿ Teaching and status presentations.
- ⦿ The six outreach simulator binaries (placeholders today).

PHYSICS COVERAGE, BY DISCIPLINE

Discipline	Crates (examples)
Nuclear data & transport	njoy-outram-park-fork, outram-mc-libs, ONIX fork
Kinetics & coupling	teh-o-prke, bedok, nee_soon, Moltres fork (precursor drift)
Thermal hydraulics & CFD	tuas_boussinesq_solver, tampines, outram-foam-* (incl. multiphase)
Properties & chemistry	tampines-steam-tables, CoolProp, DWSIM and Thermochemica forks
Fuel, structures, materials	OFFBEAT fork, farrer-park, boon-lay (TRISO)
Granular & subsurface	LIGGGHTS fork (pebble beds), PFLOTRAN fork
Severe accident & offsite	sembawang; changi (FLEXPART port, dispersion); redhill (groundwater, placeholder)
Fuel cycle	kaki-bukit (CYCLUS/CYCAMORE port, scaffold)
Twins, GUIs, tooling	digital-twin engine, dhoby-ghaut, dover, raffles, kovan, petir

Nuclear, chemical, mechanical and materials engineering all have a home.

FRONT ENDS: DOVER AND DHOBY GHAUT

	DOVER	DHOBY GHAUT
Stands for	Deck-based Open-source Visualisation Engine for Reactors	Digital High-fidelity Orchestration by GUI for a Human-friendly Automated Unified Toolkit
Fidelity For	Low: visualisation driven by input decks <i>To be decided</i> ; the 3D-engine plan pointed at a lightweight walkable plant for outreach and twins	High: drives meshing and Monte Carlo Researchers (MC Studio, Mesh Studio)
Status	Empty skeleton crate (25 Sep 2026); scope to be decided	Studios moved in, with headless modes

Neither adds physics. Both present and orchestrate what the solver crates compute.

SECTION 4

The V&V Ladder

WHEN IS A CODE “READY”?

Step	Gate	What happens
1	AI: code-to-code	Verified against its upstream <i>during</i> translation (e.g. NJOY2016 to 7 significant figures).
2	AI: test reactor	One class of test reactor per category, reproduced by an AI-generated V&V case.
3	Human V&V	Humans study the code and API, then hand-code the same cases through the user-friendly Rust API. Friction leads to API upgrades.
4	Ready	The maintainer signs off both axes in the crate's README: V&V, and human / user interface.

Passing steps 1–2 is **proof of concept**: the code is functional to a large extent and ready for human eyes. It is not a verdict.

WHAT EACH STEP PRODUCES

AI drafts (steps 1–2)

- ◉ Proof that the code runs the reactor class.
- ◉ **Publishable:** initial V&V papers, and software-engineering methodology papers on AI-drafted, human-verified scientific code.
- ◉ The signal to start human review.

Human V&V (step 3)

- ◉ Independent reproduction of the cases.
- ◉ **An API tested by use:** every friction point is a recorded fix.
- ◉ **Publishable:** the definitive V&V papers.
- ◉ Natural student projects and theses.

SECTION 5

Reactors and Test Cases

FIRST WAVE: FOUR FAMILIES, NEUTRONICS + TH/CFD

Family	Test reactor / facility	Notes
Pebble bed	HTR-10, HTTR	Double heterogeneity; HTR-10 gate exists in outtram-mc-libs
Molten salt PWR (integral)	MSRE MASLWR (Oregon State)	Precursor drift (Moltres fork) TH only: electrically heated; IAEA natural-circulation benchmark
PWR neutronics BWR	<i>open</i> <i>open</i>	e.g. BEAVRS e.g. Peach Bottom turbine trip; needs boiling / two-phase

Each family is taken up the full V&V ladder, for **both** neutronics and thermal hydraulics/CFD.

LATER WAVES: TOWARD ANY GEN IV REACTOR

- ◉ **Next: fast / liquid-metal.** EBR-I as the test reactor, pursued mainly through research collaborations.
- ◉ **Then the rest of Gen IV:** gas-cooled fast, lead-cooled, supercritical water, as collaborators and funding allow.
- ◉ Each wave reuses the ladder: code-to-code, one test reactor, human V&V, ready.

The claim to earn: *OUTRAM PARK can model any Gen IV reactor*, with a human-verified test case behind each family.

SECTION 6

Timeline

TIMELINE

When	Track	Outcome
0-12 months	AI drafting	First-wave families through AI gates 1-2; SEMBAWANG drafted
Year 1 onward	Human V&V , in parallel with more AI drafting	First-wave codes reach <i>ready</i> ; later waves drafted
Years 2-5	V&V papers	A steady flow: AI-draft papers first, then human V&V papers
Throughout	People & community	More contributors; eventually, others take over

Human V&V is the rate-limiting step. AI drafting scales; human review does not. That is why people are on this roadmap.

SECTION 7

Community

Goal: OUTRAM PARK outlives any one maintainer, as OpenFOAM does, sustained by an open community across:

Nuclear · Chemical · Mechanical · Materials
science and engineering

- ◉ **Needed in the next year or two:** more people, above all human V&V reviewers.
- ◉ **Already in place:** GPL-3.0, GOVERNANCE.md, CONTRIBUTING.md, responsible-use and provenance policies.
- ◉ **To build:** contributor onboarding, multi-institution maintainership, a user base beyond SNRSI.

HOW TO GET INVOLVED

Collaborators & researchers

- ⦿ Take a test reactor through human V&V and co-author the paper.
- ⦿ Bring a phenomenon or a Gen IV family (e.g. EBR-I).
- ⦿ Supervise student projects on the codebase.

Contributors

- ⦿ Review AI drafts; report API friction; write cases.

Funders

- ⦿ Human V&V effort: people, not compute, are the bottleneck.
- ⦿ Students and postdocs on first-wave families.
- ⦿ Collaboration on the next reactor wave.

Start here

- ⦿ github.com/theodore0nzGit/outram-park
- ⦿ github.com/theodore0nzGit/outram-park-backend

SECTION 8

Open Decisions

- ◉ **PWR neutronics test case:** BEAVRS, VERA, or other?
- ◉ **BWR test case:** Peach Bottom turbine trip, Ringhals-1 stability, or other?
- ◉ **HTR-10 and HTTR:** both, or one as the gate and one as follow-up?
- ◉ **SEMBAWANG review:** who reviews it, and what is its “ready” bar?
- ◉ **DOVER scope:** what does “deck-based visualisation” cover, and is it the outreach 3D plant?
- ◉ **Governance for handover:** what does multi-institution maintainership look like?
- ◉ **Funding targets:** which items on this roadmap are pitched to which funders?

One open codebase for anything nuclear.

Drafted with AI. Verified by people. Sustained by a community.

`github.com/theodore0nzGit/outram-park`